

NPTEL

- National Programme on Technology Enhanced Learning (NPTEL) is a project of MHRD initiated by seven Indian Institutes of Technology (Bombay, Delhi, Kanpur, Kharagpur, Madras, Guwahati and Roorkee) along with the Indian Institute of Science, Bangalore in 2003, to provide quality education to anyone interested in learning from the IITs.
- The main goal was to create web and video courses in all major branches of engineering and physical sciences at the undergraduate and postgraduate and management courses at the postgraduate level
- NPTEL is The Largest Online repository in the world of courses in engineering, Basic sciences and humanities and social science subjects
- NPTEL MOOCs - NPTEL began offering open online courses in March 2014 along with certificates from the IITs/IISc for those who completed the courses successful

Objectives Of NPTEL

The National Programme on Technology Enhanced Learning (NPTEL) was launched with the core objective of strengthening science and engineering education in India. The aim is to bring reforms that will transform India into a strong and vibrant knowledge economy.

The major objectives are:

1. Higher Education

To provide quality learning materials for undergraduate and postgraduate students in engineering, science, and technology. To ensure that students across the country, including those in remote areas, have access to the same standard of knowledge as premier institutes like the IITs and IISc.

2. Professional Education

To support continuous learning for working professionals in industries and various sectors. To help them update and upgrade their skills according to the changing demands of technology and the job market.

3. Distance Education

To create opportunities for learners who cannot attend classroom-based education in reputed institutions. To make education more flexible, accessible, and inclusive, especially for students in rural and underprivileged areas.

4. Continuous and Open Learning

To promote lifelong learning by providing open and free access to high-quality courses. To encourage self-paced learning, skill improvement, and knowledge enhancement through online platforms.

BENEFITS:

- The manpower requirement for trained engineers and technologists is much higher than the number of qualified graduates produced by Indian technical institutions. NPTEL helps in bridging this gap.
- Only a small fraction of teachers in India are fully qualified and trained; most are young, inexperienced, and hold only undergraduate degrees. NPTEL provides high-quality content to support and train teachers.
- Institutions like IITs, IISc, NITs, and leading universities use NPTEL to disseminate teaching and learning content of high quality through all available media.
- Helps in training young and inexperienced teachers so they can effectively carry out academic responsibilities. NPTEL content can be used as core curriculum material for training purposes.
- A large number of students who cannot attend premier institutions get access to quality content through NPTEL.

- Professionals and individuals in industries or other walks of life can use NPTEL for continuous training and updating knowledge, benefiting from well-developed and peer-reviewed courses by IITs and IISc.

NPTEL ONLINE CERTIFICATION

- The objective of enabling students obtain certificates for courses is to make students employable in the industry or pursue a suitable higher education programme.
- Through an online portal, 4-, 8-, or 12-week online courses, typically on topics relevant to students in all years of higher education along with basic core courses in sciences and humanities with exposure to relevant tools and technologies, are being offered.
- The enrolment to and learning from these courses involves no cost.
- Following these online courses, an in-person, proctored certification exam will be conducted and a certificate is provided through the participating institutions and industry, when applicable.
- NPTEL partners with around 3700+ colleges as on date in the form of NPTEL Local chapters